Technical Proposal — TrustGuard Fraud Detection (Team AI of Spades)

1. Problem Understanding

We predict the probability that each card/account transaction is fraudulent. Only 353 of 20,000 training transactions (1.77%) are fraud, so accuracy is meaningless: predicting "legitimate" for every row scores 98.2% accuracy while catching no fraud. We therefore optimise ranking quality under imbalance, measured by PR-AUC, and submit probabilities rather than hard labels. Operationally the score is a risk signal for step-up verification (e.g. OTP), not an automatic card block.

2. Data Processing & Feature Engineering

Checks: no duplicate rows, unique IDs (excluded from all models), valid ranges. Missing values (0.5–0.8% per column in training; 13.9% of test rows vs 4.9% of training rows have a gap) are kept, never dropped: numeric features use training-fold median imputation and categories get an explicit "missing" level. KNN and MICE imputation were tested and did not improve PR-AUC. Features: log transforms of amount, 24h spend and account age; amount relative to the 24h average transaction; night and non-SG flags; five interaction terms found by multi-factor analysis (new device × account < 90 days, new device × log amount, young account × log amount, new device × last-hour count, night × new device); and two context residuals measuring how unusual the amount and last-hour count are versus legitimate-only conditional median models, cross-fitted to avoid leakage. All preprocessing is fitted inside training folds only.

3. Model Selection & Justification

Final model: probability blend of 63% L2-regularised logistic regression (C = 0.1), 27% CatBoost (depth 5, 600 trees, strong L2) and 10% tree-augmented naive Bayes, which handles missing inputs by integrating over them. Risk factors combine roughly multiplicatively (additive in log-odds), so logistic regression was the strongest single model; the tree and Bayesian components add thresholds and feature dependencies. Over a dozen alternatives (XGBoost, LightGBM, SVM, TabPFN, AutoGluon, EBM, neural tabular models, clustering and outlier features) scored lower or did not add value. Class weighting and resampling did not improve PR-AUC, so the natural class prior is kept.

4. Evaluation Strategy

Stratified 5-fold cross-validation, repeated over five random splits, with every preprocessing step and donor model refitted within each training fold. Primary metric: out-of-fold average precision (PR-AUC). The final blend scores 0.2365 on a held-out 5-fold split (constant baseline 0.0177; plain logistic regression ≈ 0.231) and 0.233 averaged over 25 repeated folds. Each experiment's adoption rule was written before results were seen, to limit selection bias. Public leaderboard PR-AUC: 0.18900 (best of our submissions). The package was verified to reproduce its predictions exactly in a fresh process.

5. Limitations & Risks

No customer identifier or timestamp exists, so per-customer history — the main source of gains in published fraud systems — cannot be built; most missed frauds look like ordinary known-device transactions. With 353 positives, validation is noisy and repeatedly reused labels risk optimistic model selection. The test set is shifted (more missing values, more new-device and high-activity rows; adversarial AUC 0.66), and local gains did not always transfer to the public score (CV ≈ 0.24 vs public ≈ 0.19). Fraud patterns drift, so the model and alert threshold should be monitored and retrained regularly.